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Computational HW#2
2/10/26

1. Consider

$$x^3 - 7x^2 + 14x - 5 = 0$$

Find soln by hand to a precision in the value of x smaller than 0.01.

a) Using the bisection method w/initial guesses of $x_0 = 0$, $x_1 = 1$

$x_2 = \frac{x_1 + x_0}{2}$ "General" formula,

$f(x_0) = 0 - 0 + 0 - 5 = -5$

$f(x_1) = 1 - 7 + 14 - 5 = 3$

$x_2 = \frac{1+0}{2} = 1/2 = 0.5$

$f(x_2) = (\frac{1}{2})^3 - 7(\frac{1}{2})^2 + 14(\frac{1}{2}) - 5$

$= \frac{1}{8} - \frac{7}{4} + 7 - 5$

$= -\frac{13}{8} + 2 = \frac{3}{8} = 0.375$

$x_3 = \frac{0 + 1/2}{2} = \frac{1}{4} = .25$

$f(x_3) = (\frac{1}{4})^3 - 7(\frac{1}{4})^2 + 14(\frac{1}{4}) - 5$

$= \frac{1}{64} - \frac{7}{16} + \frac{7}{2} - 5$

$= \frac{-123}{64} = -1.922$

$x_4 = \frac{\frac{1}{2} + \frac{1}{4}}{2} = \frac{\frac{3}{4}}{2} = \frac{3}{8} = 0.375$

$f(x_4) = (\frac{3}{8})^3 - 7(\frac{3}{8})^2 + 14(\frac{3}{8}) - 5$

$= \frac{27}{512} - 7(\frac{9}{64}) + \frac{42}{8} - 5$

$= \frac{-349}{512} = -0.682$

$x_5 = \frac{\frac{3}{8} + \frac{1}{2}}{2} = \frac{\frac{7}{8}}{2} = \frac{7}{16}$

$f(x_5) = (\frac{7}{16})^3 - 7(\frac{7}{16})^2 + 14(\frac{7}{16}) - 5 = -0.131$

$x_6 = \frac{\frac{7}{16} + \frac{1}{2}}{2} = \frac{15}{32}$

$f(x_6) = (\frac{15}{32})^3 - 7(\frac{15}{32})^2 + 14(\frac{15}{32}) - 5 = \frac{4175}{32768} = 0.127$

$x_7 = \frac{7/16 + 15/32}{2} = \frac{29}{64}$

$f(x_7) = (\frac{29}{64})^3 - 7(\frac{29}{64})^2 + 14(\frac{29}{64}) - 5 = -4.69 \times 10^{-4}$

X_n	x (frac)	x (decimal)	$f(x_n)$
x_0	0	0	-5
x_1	1	1	3
x_2	1/2	0.5	0.375
x_3	1/4	.25	-1.922
x_4	3/8	0.375	-0.682
x_5	7/16	0.4375	-0.131
x_6	15/32	0.46875	0.127
? → x_7	29/64	0.453	-4.69×10^{-4}
x_8	59/128	0.4609	0.064
x_9	117/256	0.457	0.032
x_{10}	233/512	0.455	0.016
x_{11}	465/1024	0.454	0.0076
x_{12}	929/2048	0.4536	0.00357
x_{13}	1857/4096	0.4534	0.00155
x_{14}	3713/8192	0.45325	5.405×10^{-4}
x_{15}	7425/16384	0.45319	3.567×10^{-5}

$x_8 = \frac{29/64 + 15/32}{2} = \frac{59}{128}$

$f(x_8) = (\frac{59}{128})^3 - 7(\frac{59}{128})^2 + 14(\frac{59}{128}) - 5 = 0.064$

$x_9 = \frac{29/64 + 59/128}{2} = \frac{117}{256}$

$f(x_9) = (\frac{117}{256})^3 - 7(\frac{117}{256})^2 + 14(\frac{117}{256}) - 5 = 0.032$

$x_{10} = \frac{29/64 + 117/256}{2} = \frac{233}{512}$

$f(x_{10}) = (\frac{233}{512})^3 - 7(\frac{233}{512})^2 + 14(\frac{233}{512}) - 5 = 0.016$

$x_{11} = \frac{29/64 + 233/512}{2} = \frac{465}{1024}$

$f(x_{11}) = (\frac{465}{1024})^3 - 7(\frac{465}{1024})^2 + 14(\frac{465}{1024}) - 5 = 0.0076$

$x_{12} = \frac{29/64 + 465/1024}{2} = \frac{929}{2048}$

$f(x_{12}) = (\frac{929}{2048})^3 - 7(\frac{929}{2048})^2 + 14(\frac{929}{2048}) - 5 = 0.00357$

$x_{13} = \frac{29/64 + 929/2048}{2} = \frac{1857}{4096}$

$f(x_{13}) = (\frac{1857}{4096})^3 - 7(\frac{1857}{4096})^2 + 14(\frac{1857}{4096}) - 5 = 0.00155$

$x_{14} = \frac{29/64 + 1857/4096}{2} = \frac{3713}{8192}$

$f(x_{14}) = (\frac{3713}{8192})^3 - 7(\frac{3713}{8192})^2 + 14(\frac{3713}{8192}) - 5 = 5.405 \times 10^{-4}$

$x_{15} = \frac{29/64 + 3713/8192}{2} = \frac{7425}{16384}$

$f(x_{15}) = (\frac{7425}{16384})^3 - 7(\frac{7425}{16384})^2 + 14(\frac{7425}{16384}) - 5 = 3.567 \times 10^{-5}$